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Tank

Abstract

A tank includes: a wall separating a liquid storage chamber, which stores a liquid to be supplied to a liquid consuming part, from an outside; an inlet penetrating the wall; and a liquid outflow part which allows the liquid to flow to the outside of the liquid storage chamber. The wall includes a view wall, which extends upward and downward and allows a user to visually check a liquid level in the liquid storage chamber, and a bottom wall, which defines an inner bottom surface of the liquid storage chamber by being connected to the view wall. The inner bottom surface of the bottom wall includes an inclined part which extends from at least a part of the view wall to incline downward in a direction away from the view wall.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 17/062,829, filed Oct. 5, 2020, now U.S. Pat. No. 11,396,186, which is a continuation of U.S. patent application Ser. No. 16/543,206, filed Aug. 16, 2019, now U.S. Pat. No. 10,792,928, which is a continuation of U.S. patent application Ser. No. 15/952,662, filed Apr. 13, 2018, now U.S. Pat. No. 10,384,455, which is a continuation of U.S. patent application Ser. No. 15/595,238, filed May 15, 2017, now U.S. Pat. No. 9,944,085, which is a continuation of U.S. patent application Ser. No. 14/996,617, filed Jan. 15, 2016, now U.S. Pat. No. 9,649,844, all of which further claim priority from Japanese Patent Application No. 2015-008171 filed on Jan. 19, 2015, the disclosure of all of which are incorporated herein by reference in their entirety.

BACKGROUND

Field of the Invention

(1) The present invention relates to a tank used for a liquid consuming part of an ink-jet printer or the like.

Description of the Related Art

(2) There has been conventionally known, as an exemplary liquid consuming apparatus, a printer including a capacious tank which can be replenished with ink and a recording head which discharges the ink supplied from the tank through nozzles to record letters, an image, and the like on a recording sheet. The tank includes an ink inlet which is opened and closed by a cap. Removing

the cap from the inlet allows the tank to be refilled with the ink through the inlet. The tank includes a view wall with an alarm line (a replenishment-start line) for alerting a shortage of the ink to a user, the view wall being a vertical wall used for checking a remaining amount of ink in the tank. A user refills the tank with the ink when a liquid level of ink in the tank fails to reach a position indicated by the replenishment-start line.

SUMMARY

(3) The replenishment-start line, however, is provided at a lower part of the view wall near the bottom surface of a liquid chamber of the tank. Thus, when ink is remained at a corner between an inner surface of the view wall and the bottom surface of the liquid chamber by surface tension in a state that the liquid amount of ink in the tank is substantially zero, a liquid level of the ink may look as if it were above the replenish start line. As a result, the user can not know that the liquid amount of ink is substantially zero, and thus the user may miss the timing to refill the tank with the ink.

(4) Missing the ink supply timing causes air to enter a tube through which the ink is supplied from the tank to the recording head. This may cause printing failure, temporary interruption of printing, or the like.

(5) In view of the above, an object of the present teaching is to provide a tank which allows a user to know a remaining amount of liquid in the tank at appropriate timing.

(6) According to an aspect of the present teaching, there is provided a tank configured to store a liquid, including: a liquid storage chamber; a wall defining the liquid storage chamber, the wall including a view wall and a bottom wall, the view wall having transparency or translucency so that a liquid level in the liquid storage chamber is visible, the view wall extending in an up-down direction, and the bottom wall being connected to the view wall to define an inner bottom surface of the liquid storage chamber; an inlet penetrating the wall; and a liquid outflow part configured to allow the liquid to flow to the outside of the liquid storage chamber, wherein the inner bottom surface of the bottom wall includes an inclined surface which is formed in at least a part of the bottom wall connected to the view wall, and which extends downward in a direction away from the view wall; and the liquid outflow part is positioned below a boundary between the inclined surface and the view wall.

(7) In this configuration, when the liquid is consumed in a state that the liquid level in the liquid storage chamber fails to reach the upper end of the inclined surface connected to the view wall, the liquid moves or flows downward in the direction away from the view wall. Thus, the liquid is prevented from remaining at the inner surface of the view wall, and the user can know the decrease in liquid at appropriate timing. Accordingly, the user can perform liquid replenishment at appropriate timing.

(8) The present teaching can prevent the liquid from remaining at the inner surface of the view wall when the liquid in the liquid storage chamber is consumed or reduced. This allows the user to appropriately know the decrease in liquid, thereby making it possible to stably replenish the liquid storage chamber with the liquid at appropriate timing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. 1A and 1B are exemplary schematic views of an ink-jet printer including a tank according to an embodiment of the present teaching, wherein FIG. 1A depicts a state in which a cover is closed, and FIG. 1B depicts a state in which the cover is open.

(2) FIG. 2 is a vertical cross-sectional view schematically depicting an internal structure of a printer unit.

(3) FIG. 3 is a front view of the tank according to the embodiment of the present teaching.

- (4) FIG. 4 is a vertical cross-sectional view depicting a state in which the cover is open and a cap closes the inlet of the tank.
- (5) FIG. 5 is a vertical cross-sectional view depicting a state in which the cover is open and the cap is removed from the inlet of the tank.
- (6) FIG. 6 is a cross-sectional view taken along the arrow VI-VI in FIG. 3.
- (7) FIG. 7 is an enlarged cross-sectional view of an area around an inclined part depicted in FIG. 6.
- (8) FIG. 8 is a cross-sectional view taken along the arrow VIII-VIII in FIG. 3.
- (9) FIG. 9 is a cross-sectional view taken along the arrow IX-IX in FIG. 8.

DESCRIPTION OF THE EMBODIMENTS

(10) An explanation will be made about an embodiment of the present teaching with reference to drawings. In the embodiment, a liquid consuming apparatus is exemplified by an ink-jet multifunction printer **10** (hereinafter also referred to simply as “multifunction peripheral **10**”). The multifunction peripheral **10** in this embodiment has various functions such as a facsimile function and a print function. In this specification and claims, an up-down direction is coincident with an up-down direction **7** of the multifunction peripheral **10** depicted in FIG. 1. Similarly, a front-rear direction and a left-right direction in the specification and claims are respectively coincident with a front-rear direction **8** and a left-right direction **9** of the multifunction peripheral **10** depicted in FIG. 1. The embodiment to be explained below is merely an example of the present teaching, and it is needless to say that the embodiment of the present teaching can be changed or modified appropriately without departing from the gist and scope of the present teaching.

(11) <Overall Structure of Multifunction Peripheral **10**>

(12) As depicted in FIGS. 1A and 1B, the multifunction peripheral **10** has a substantially rectangular parallelepiped form. The multifunction peripheral **10** includes a printer unit **11** in its lower part. The printer unit **11** records letters, an image, and the like on a sheet **12** (see FIG. 2) in accordance with an ink-jet recording system. An opening **13** is provided in a front wall **14A** of a housing **14** of the printer unit **11** in the center in the left-right direction **9**. An operation panel **17** is provided above the opening **13**. The operation panel **17** includes input buttons **17A** and a liquid crystal display **17B**.

(13) The opening **13** includes a feed tray **20** and a discharge tray **21**. A user inserts or removes the feed tray **20** into or from the multifunction peripheral **10** in the front-rear direction **8**. The feed tray **20** accommodates sheets **12** stacked thereon. The discharge tray **21** is positioned above the feed tray **20**.

(14) A tank **100** storing inks is provided in a right front part of the housing **14**. The tank **100**, which is accommodated in the housing **14**, is fixed to the housing **14** so that no user can easily remove the tank **100** from the housing **14**.

(15) The front surface of the tank **100** is exposed to the outside of the multifunction peripheral **10** via an opening **22**, which is formed in the front wall **14A** of the housing **14**. The housing **14** is provided with a cover **70** which is swingable between a closed position (see FIG. 1A) where the cover **70** covers the opening **22** and an open position (see FIG. 1B) where the opening **22** is exposed. The external form and size of the cover **70** correspond to the opening **22**.

(16) The cover **70** is supported by the housing **14** so that a lower part of the cover **70** is swingable around a swing axis **70A** extending in the left-right direction **9**. The cover **70** in the closed position covers the front part of the tank **100** to form a part of the front wall **14A** of the housing **14**. The cover **70** in the open position exposes the front part of the tank **100** to the outside of the housing **14**.

(17) As depicted in FIG. 1A, the cover **70** includes a window **74** in the center of its front surface. The window **74** allows light to pass therethrough. The window **74** is formed, for example, of a transparent material which is placed or embedded in the opening to make visible light pass. The window **74** has such a size that the user can visually check the front part of the tank **100** in the up-down direction **7**.

(18) The window **74** may not be formed of the transparent material allowing visible light to pass.

For example, the window **74** may be formed only of the opening. In such a case, it is preferred that the window **74** have such a size that the user, who sees the front part of the tank **100** from the outside with the cover **70** closed, can check a remaining amount of ink, as will be described later.

(19) <Printer Unit **11**>

(20) As depicted in FIG. 2, the printer unit **11** includes the feed tray **20** accommodating sheets **12**, a feed unit **15** feeding the sheet **12** in the feed tray **20**, a conveyance roller unit **54** feeding the sheet **12** fed from the feed unit **15** to a recording unit (liquid consuming unit) **24**, the recording unit **24** recording letters, an image, and the like on the sheet **12**, a discharge roller unit **55** discharging the sheet **12** for which recording has been performed by the recording unit **24**, and the discharge tray **21** receiving the sheet **12** discharged from the discharge roller unit **55**.

(21) The feed unit **15** includes a feed roller **25**, a feed arm **26**, and a shaft **27**. The front end of the feed arm **26** rotatably supports the feed roller **25**. A conveyance motor (not depicted) rotates the feed roller **25** to feed the sheet **12** in a sheet conveyance direction **16**. The shaft **27**, which is supported by a frame of the printer unit **11**, pivotably or swingably supports the feed arm **26**. An elastic force caused by a spring or a weight of the feed arm **26** biases the feed arm **26** so that the feed arm **26** swings toward the feed tray **20**. The feed unit **15** feeds each sheet **12** accommodated in the feed tray **20** to a conveyance path **65**.

(22) The conveyance path **65** extends from the rear end of the feed tray **20** toward the rear side of the printer unit **11**, extends from the lower side to the upper side in the up-down direction **7** on the rear side of the printer unit **11** while being curved to make a U-turn, and passes through the space between the recording unit **24** and the platen **42** to arrive at the discharge tray **21**. The rear end of the feed tray **20** and a space formed by an outer guide member **18** and an inner guider member **19** which face each other while being separated by a predetermined interval constitute a part of the conveyance path **65** through which the sheet **12** is conveyed to the recording unit **24**. The arrows using dashed-dotted lines in FIG. 2 indicate the sheet conveyance direction **16** of the sheet **12** in the conveyance path **65**. The conveyance path **65** is provided in the center of the multifunction peripheral **10** in the left-right direction **9**.

(23) The conveyance roller unit **54** is disposed on the upstream side of the recording unit **24** in the sheet conveyance direction **16**. The conveyance roller unit **54** includes a conveyance roller **60** and a pinch roller **61** facing each other. The conveyance roller **60** is driven by an unillustrated conveyance motor. The pinch roller **61** is driven to rotate accompanying with the rotation of the conveyance roller **60**. The sheet **12** is conveyed in the sheet conveyance direction **16** while being held or nipped by the conveyance roller **60** and the pinch roller **61**.

(24) The recording unit **24** includes a carriage **23** and the platen **42** facing each other in the up-down direction **7** with the conveyance path **65** intervening therebetween. The carriage **23** includes the recoding head **39**. The recording head **39** includes nozzles **40** in its lower surface. Ink is supplied from the tank **100** to the recording head **39**. The recording head **39** discharges the ink as fine ink droplets through the nozzles **40**. Namely, the recording head **39** discharges ink droplets onto the sheet **12**, which is being supported by the platen **42** from the lower side, during movement of the carriage **23**. Accordingly, letters, an image, and the like are recorded on the sheet **12**.

(25) The discharge roller unit **55** is disposed on the downstream side of the recording unit **24** in the sheet conveyance direction **16**. The discharge roller unit **55** includes a discharge roller **62** and a spur roller **63** facing each other. The discharge roller **62** is driven by a conveyance motor (not depicted). The spur roller **63** is driven to rotate accompanying with the rotation of the discharge roller **62**. The sheet **12** is conveyed in the sheet conveyance direction **16** while being held or nipped by the discharge roller **62** and the spur roller **63**. The sheet **12**, which has passed through the discharge roller unit **55**, is discharged on the upper part of the discharge tray **21**.

(26) <Tank **100**>

(27) As depicted in FIGS. 1A, 1B, and FIG. 3, the tank **100** has a substantially rectangular parallelepiped form. The tank **100** includes walls separating ink chambers **111** from the outside.

The walls include a view wall **101** as its front surface. The view wall **101** extends upward and downward to allow the user to visually check a liquid level of each ink. Further, the walls include a second side wall **103** connected to the view wall **101** to define a side surface which extends in a direction intersecting with the view wall **101**; an upper wall **104** connected to the view wall **101** to define the upper surface of the ink chambers **111**; and a bottom wall **105** defining an inner bottom surface **105A** (FIGS. **6** and **8**) of the ink chambers **111**.

(28) The internal space of the tank **100** according to this embodiment is divided into four spaces in the left-right direction **9** to define four ink chambers **111B**, **111M**, **111C**, and **111Y** disposed adjacently to each other. Each of the four ink chambers **111B**, **111M**, **111C**, and **111Y** is an exemplary liquid storage chamber. The alphabetic suffixes of B, M, C, and Y show ink colors respectively. The ink chambers **111B**, **111M**, **111C**, and **111Y** will be described collectively as “ink chambers **111**” in some cases.

(29) First side walls **102** partition between the ink chambers **111B** and **111M**, between the ink chambers **111M** and **111C**, and between the ink chambers **111C** and **111Y**, respectively. This configuration forms the spaces of the ink chambers **111B**, **111M**, **111C**, and **111Y**. The rear surface of the tank **100** is sealed (closed) by welding a film **120** (FIG. **4**) to the rear end surfaces of the first side walls **102**, the second side wall **103**, the upper wall **104**, and the bottom wall **105**.

(30) The ink chamber **111B** contains black ink, the ink chamber **111M** contains magenta ink, the ink chamber **111C** contains cyan ink, and the ink chamber **111Y** contains yellow ink. Each of the inks is an exemplary liquid. The number of ink chambers **111** and ink colors are not limited to the above examples. Of the four ink chambers **111B**, **111M**, **111C**, and **111Y**, the ink chamber **111B** is disposed on the right most side, and the ink chamber **111Y** is disposed on the left most side. The ink chamber **111B** has a capacity larger than those of other ink chambers **111M**, **111C**, and **111Y**.

(31) An inclined wall **101B** includes the corners of the view wall **101** and the upper wall **104**. Further, the inclined wall **101B** includes inlets **112B**, **112M**, **112C**, and **112Y** through which the inks are poured into the ink chambers **111B**, **111M**, **111C**, and **111Y** respectively. The inlets **112B**, **112M**, **112C**, and **112Y** will be described collectively as “inlets **112**” in some cases.

(32) The inlets **112** penetrate the inclined wall **101B** of the tank **100** in its thickness direction to allow the ink chambers **111** corresponding thereto respectively to communicate with the outside of the tank **100**. In this embodiment, the inlets **112** are formed in the inclined wall **101B** of the tank **100**. The inlets **112** face outward and obliquely upward when fixed to the housing **14**. Namely, the inlets **112** open into the inclined wall **101B** in a direction parallel to an axis L (FIG. **5**), which inclines to the up-down direction **7** and the front-rear direction **8**.

(33) In this embodiment, the view wall **101** of the tank **100** includes an ink replenishment line **110**. The ink replenishment line **110** lets the user know the timing for replenishing the tank **100** with ink. For example, the ink replenishment line **110** can be provided at a position indicating that a remaining amount of ink in each ink chamber **111** is 15-25% of the capacity of each ink chamber **111**. Alternatively, the ink replenish line **110** of the tank **100** is positioned, for example, at the same height as a position where the view wall **101** is connected to an inclined part **106** as will be described later. The ink replenish line **110** may be a mark or a seal which is put on the view wall **101** of the tank **100** to indicate the ink replenish line. The ink replenishment line **110** may be formed in the cover **70** of the multifunction peripheral **10**. The view wall **101** also includes an ink full-amount line **109** in its upper part. The colors of inks and the like may be indicated, for example, by putting colored marks or seals on the tank **100** or indicating colors of inks on the cover **70**.

(34) As depicted in FIG. **1B**, the inlets **112** of the tank **100** are exposed to the outside of the multifunction peripheral **10** through the opening **22** in a state that the cover **70** is in the open position. Removable Caps **113B**, **113M**, **113C**, and **113Y** are respectively provided to the inlets **112B**, **112M**, **112C**, and **112Y** (FIG. **3**). The caps **113B**, **113M**, **113C**, and **113Y** will be referred to collectively as “caps **113**” in some cases. Removing the cap **113** from the inlet **112** in a state that

the cover **70** supported by the housing **14** is in the open state opens the inlet **112**. The ink can be poured into the ink chamber **111** through the inlet **112**.

(35) The tank **100** according to this embodiment is formed of a translucent or transparent resin. The inlets **112** are provided in the inclined wall **101B**, which is an upper part of the view wall **101**. Thus, the user can replenish the tank **100** with the ink from the inlet **112** while visually checking a liquid level through the view wall **101**. This enables the user to easily replenish the tank **100** with an appropriate amount of ink.

(36) <Cap **113**>

(37) As depicted in FIG. 4, the cap **113** fitted in the inlet **112** closes the inlet **112** by being brought into tight contact therewith. The caps **113B**, **113M**, **113C**, and **113Y** have the same shape. In the following, detailed configuration will be explained by using the collective term “caps **113**”.

(38) The caps **113** each include a first part **115** and a second part **116**. The first part **115**, which has a columnar shape, is inserted into a path **114** of the inlet **112** to close the path **114**. The second part **116**, which has a large-diameter columnar shape, is formed continuously to the first part **115** and is positioned outside the inlet **112**. The second part **116** includes an arm **117** having a first end connected to a part, of the second part **116**, which is deviated from the center of the second part **116**. A second end of the arm **117** is inserted or supported between an upper part of the tank **100** and the housing **14**. Thus, the cap **113** is fixed between the tank **100** and the housing **14** in a state of being prevented from being removed easily. The cap **113** is formed of an elastically deformable material such as rubber and elastomer.

(39) Closing the inlet **112** with the first part **115** of the cap **113** as depicted in FIG. 4 causes the arm **117** to bend or curve upward, that is, to have a convex-like shape. When the ink in each ink chamber **111** of the tank **100** becomes insufficient by being consumed, the user pivotably moves the cover **70** from the closed position to the open position so as to access the inlet **112** of the tank **100** through the opening **22** of the housing **14** (the state depicted in FIG. 4).

(40) Then, the user removes the first part **115** and the second part **116** of the cap **113** from the inlet **112**, as depicted in FIG. 5. The first part **115** and the second part **116** removed from the inlet **112** stand and extend upward by the aid of the elastic force of the arm **117**, thus forming a space around the inlet **112**. Opening the inlet **112** in such a manner enables the user to replenish the ink chamber **111** with the ink through the inlet **112**.

(41) When completing the ink replenishment, the user seals the inlet **112** by inserting the first part **115** of the cap **113** into the inlet **112** (the state depicted in FIG. 4). After that, the user pivotably moves the cover **70** from the open position to the closed position (the state depicted in FIG. 1A).

(42) <Ink Chambers **111**>

(43) Subsequently, the ink chambers **111** will be explained with reference to FIG. 6. FIG. 6 depicts a cross-section of the ink chamber **111Y** having the same internal space as those of the ink chambers **111M** and **111C**. The ink chambers **111M**, **111C**, and **111Y** have the same structure, and thus any explanation of the ink chambers **111M** and **111C** will be omitted.

(44) The ink chamber **111Y** is a space defined by the view wall **101**, the upper wall **104**, the bottom wall **105**, the first side wall **102** and second side wall **103** which extend in a direction intersecting with these walls **101**, **104**, and **105** to define side surfaces (FIG. 3), and the film **120** to be putted on the rear end surface of the ink chamber **111Y**. The example depicted in FIG. 6 includes a buffer wall **108** which extends in a horizontal direction at an intermediate part in the up-down direction **7**. The ink poured from the inlet **112** formed in the inclined wall **101B** falls onto the buffer wall **108** to reduce the speed of ink current or flow, and then falls downward from buffer wall **108** in the ink chamber **111Y**.

(45) An ink outflow part **118Y** through which the ink flows is formed at the rear lower end of the ink chamber **111Y**. The ink outflow part **118Y** extends to the outer surface of the second side wall **103** disposed on the right end of the tank **100**. The ink chambers **111B**, **111M**, and **111C** include ink outflow parts **118B**, **118M**, and **118C** through which the inks stored in the ink chambers **111B**,

111M, and **111C** flow from the tank **100** to the recording unit **24**.

(46) The inclined part **106**, which is connected to the inner surface of the view wall **101**, is a part of the inner bottom surface **105A** of the bottom wall **105** of the ink chamber **111Y**. The inclined part **106** extends from the connection point with the inner surface of the view wall **101** to incline downward and rearward. In other words, the inclined part **106** of the bottom wall **105** forms an obtuse angle with the inner surface of the view wall **101** extending in the up-down direction **7**. A lowermost part **105B** horizontally extends rearward from the rear end of the inclined part **106**. Namely, the inclined part **106** of the inner bottom surface **105A** according to this embodiment extends from the inner surface of the view wall **101** to the lowermost part **105B**. Similar to the ink chamber **111Y**, each of the ink chambers **111M** and **111C** includes the bottom wall **105** having the inner bottom surface **105A** provided with the inclined part **106** and the lowermost part **105B**. Although a step between the inclined part **106** and the lowermost part **105B** is formed in this embodiment, the inclined part **106** may be formed continuously to the lowermost part **105B** without the step.

(47) As depicted in FIG. 7, the view wall **101** includes the ink replenishment line **110** on its outer surface. The inclined part **106**, which is connected to the inner surface of the view wall **101** at a position at which the ink replenishment line **110** is provided on the outer surface, extends from the connection point to incline rearward and downward. The inclined part **106** according to this embodiment is formed to have an inclination angle θ of approximately 20 to 25 degrees relative to a horizontal plane.

(48) The inclined part **106**, which has an inclination angle θ of at least not less than two degrees relative to the horizontal plane, prevents the ink from remaining at a corner connecting the inner surface of the view wall **101** and the inclined part **106**, when the ink is consumed. It is preferred that the inclination angle θ of the inclined part **106** be not less than 10 degrees. The inclined part **106**, which has an inclination angle θ of not less than 10 degrees relative to the horizontal plane, further prevents the ink from

(49) remaining at the corner. This enables the user to appropriately know that a liquid level of the ink is away from the inner surface of the view wall **101**. Even when the multifunction peripheral **10** is set in a slightly inclined state (for example, in a state of being inclined forward), the inclined part **106** prevents the ink from remaining at the corner and enables the user to appropriately know the decrease in ink.

(50) Subsequently, the ink chamber **111B** will be explained with reference to FIG. 8. The ink chamber **111B** includes the buffer wall **108** which extends in the horizontal direction at an intermediate part in the up-down direction **7**. The ink poured through the inlet **112** formed in the inclined wall **101B** falls onto the buffer wall **108** to reduce the speed of ink current or flow, and then falls downward from buffer wall **108** in the ink chamber **111B**.

(51) Of the four ink chambers, the ink chamber **111B** is positioned on the outermost side in the tank **100**. The ink chamber **111B** has an inclined part different from those of the ink chambers **111Y**, **111M**, and **111C**. In the tank **100** according to this embodiment, the ink outflow parts **118Y**, **118C**, and **118M** of the ink chambers **111Y**, **111C**, and **111M** disposed on the left side of the ink chamber **111B** extend in the left-right direction **8** below the ink chamber **111B**. Especially, the ink outflow part **118M** of the ink chamber **111M** is formed on the lower front side of the ink chamber **111B**. Thus, the bottom wall **105** of the ink chamber **111B** is formed of an inner bottom surface **105A** provided with a lowermost part **105B**, an inclined part **106**, and a step **105C**. The lowermost part **105B** is formed at a position separated from the view wall **101**. The inclined part **106** extends from the inner surface of the view wall **101** to incline downward and rearward. The step **105C** connects the inclined part **106** and the lowermost part **105B**. The lowermost part **105B** is disposed below the inclined part **106**. Since the ink outflow part **118M** is formed on the lower side of the ink chamber **111B**, the inclined part **106** has a gentle inclination angle θ , for example, about 2 to 5 degrees. The space for the ink outflow part **118M** extending from the ink chamber **111M** can be made below the

inclined part **106** of the ink chamber **111B** of the tank **100** by making the inclination angle θ of the inclined part **106** gentle and providing the step **105C** which connects the inclined part **106** and the lowermost part **105B**. The ink outflow part **118B** of the ink chamber **111B** is provided at a position between the inclined part **106** and the lowermost part **105B** in the up-down direction **7**.

(52) The inclined part **106** includes grooves **107** extending downward from the connection point with the view wall **101** to the step **105C**. An end, of each groove **107**, on the opposite side to the view wall **101** is open. The grooves **107** generate capillarity, which allows the ink to flow from the inclined part **106** to the step **105C** when the ink is consumed, so that no ink remains at the boundary between the view wall **101** and the inclined part **106**. Thus, the inclined part **106** having a gentle inclination angle θ can prevent the ink from remaining at the inner surface of the view wall **101**. The capillarity generated by the grooves **107** allows the ink to fall onto the lowermost part **105B**, thereby making it possible to efficiently concentrate or gather the ink in the ink outflow part **118B** disposed between the inclined part **106** and the lowermost part **105B** in the up-down direction **7**. In order to move the ink toward the lowermost part **105B** by the capillarity, it is preferred that the width of each groove **107** be about 0.5 to 1 mm.

(53) As depicted in FIG. **9**, the grooves **107** are formed at a part, of the ink chamber **111B**, close to the second side wall **103**. Namely, the grooves **107** are formed at a part (right end in this example), of the inclined part **106**, closer to the second side wall **103** than to the center position in a direction intersecting with the second side wall **103**.

(54) Accordingly, the ink is prevented from remaining at the inner surface of the view wall **101** of the tank **100** including the ink chambers **111B**, **111M**, **111C**, and **111Y** and being configured so that the ink outflow parts **118M**, **118C**, and **118Y** of the ink chambers **111M**, **111C**, and **111Y** pass below the ink chamber **111B**. Specifically, the inner bottom surface **105A** of the ink chamber **111B** is required to have the inclined part **106** and the step **105C** to provide the space for the ink outflow part **118M** (FIG. **8**). The inclined part **106**, however, may not be formed over the entire view wall **101** of the ink chamber **111B** in the left-right direction **9**. In such a case also, the ink is prevented from remaining at the corner of the inner surface of the view wall **101** by providing the inclined part **106** at the position close to the second side wall **103** and providing the grooves **107** in the inclined part **106**.

(55) Thus, even the inclined part **106** having a gentle inclination angle θ allows the ink to move or flow downward and to be prevented from remaining at the inner surface of the view wall **101**. This is because, the capillarity of the grooves **7** formed near the second side wall **103** works, when the ink in the ink chamber **111B** fails to reach the corner formed by the inclined part **106** of the inner bottom surface **105A** and the view wall **101**. The user can appropriately know that a remaining ink fails to reach the position indicated by the ink replenishment line **110**, accordingly.

(56) <Technical Effect of the Embodiment>

(57) According to the tank **100** of this embodiment, when the ink (liquid) stored in the ink chamber (liquid storage chamber) **111** is consumed in a state that the liquid level of ink (liquid) fails to reach the upper end of the inclined part **106** connected to the view wall **101**, the ink moves or flows downward along the inclined part **106** in the direction away from the view wall **101**. Thus, the tank **100** prevents the ink from remaining at the inner surface of the view wall **101** and enables the user to know the decrease in ink at appropriate timing. Accordingly, the user can replenish the tank **100** with the ink at appropriate timing.

(58) The tank **100** according to this embodiment prevents printing failure and temporary interruption of printing which would be otherwise caused by the decrease in ink. Namely, the tank **100** according to this embodiment results in stable printing.

OTHER EMBODIMENTS

(59) In the above embodiment, the ink is used as an example of liquid. The present teaching, however, is not limited to this. Namely, instead of the ink, the liquid may be a pretreatment liquid which is to be discharged onto a recording sheet before the ink to be jetted at the time of printing,

or may be water or the like which is to be sprayed in the vicinity of the nozzles **40** of the recording head **39** for preventing drying of the nozzles **40** of the recording head **39**.

(60) In the above embodiment, the tank **100** storing inks of four colors is used as an example. The present teaching, however, is not limited to this. It is possible to use a tank storing a single kind of liquid or a tank storing more kinds of liquids. Such tanks may include the liquid storage chamber(s) corresponding to the number of kinds of liquids.

(61) The shape of the cap **113** may be changed appropriately. For example, instead of the shape insertable into the inlet **112** of the tank **100**, the cap **113** may have a shape to be fitted onto a cylindrical projection formed around the inlet **112**. Or, the cap **113** may be configured to seal the inlet **112** in such a manner that a male screw formed around the inlet **112** is screwed into a female screw formed in the cap **113**.

(62) In the above embodiment, the opening **22** is formed on the right side of the front wall **14A** of the housing **14** and the tank **100** is disposed on the rear side of the opening **22**. The opening **22**, however, may be formed on the left side of the front wall **14A** and the tank **100** may be disposed on the rear side of the opening **22**. Or, instead of providing the opening **22** in the front wall **14A** of the housing **14**, the opening **22** may be formed in the right lateral wall or the left lateral wall so that a user can access the inlets **112** of the tank **100** from the right side or the left side.

Claims

1. An ink-jet printer comprising: a casing having a casing front wall, wherein the casing front wall has a first opening and a second opening; a feed tray disposed in the first opening; and a refillable tank disposed in the casing, wherein the refillable tank is fixed so as not to be removed from the second opening, the refillable tank including: an upper ink chamber; a lower ink chamber; a wall defining the upper ink chamber and the lower ink chamber, the wall comprising a front wall, a rear wall, first side wall and second side wall, the front wall having transparency or translucency, the front and the rear walls extending in an up-down direction and a left-right direction, the first and second side walls extending in the up-down direction and a front-rear direction, wherein an area of the front wall is smaller than an area of each of the side walls, the front wall is exposed from the second opening and the front wall includes a lower limit indicator of the ink; and another wall which divides the upper ink chamber and the lower ink chamber and which extends rearward from the front wall at a position above the lower limit indicator of the ink in the up-down direction, wherein the upper ink chamber communicates with the lower ink chamber through a communication port formed by the another wall, the communication port being located rearward of an inlet for refilling the tank, and the inlet being accessible through the second opening, and wherein, when viewed from the up-down direction, the inlet through which the ink is injected to refill the tank does not overlap the communication port.
 2. The ink-jet printer according to claim 1, wherein, when viewed from the up down direction, the communication port does not overlap with an inlet through which ink is injected.
 3. The ink-jet printer according to claim 1, wherein the ink injected from an inlet falls on the another wall and then falls from the upper ink chamber to the lower ink chamber.
 4. The ink-jet printer according to claim 1, wherein the communication port is separated from the front wall.
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